

Boğaziçi University
Institute for Data Science & Artificial Intelligence

DSAI 512 Fall 2025

HW #5

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Due: 14/12/2025

General submission information:

- Each homework must be submitted as a single `.pdf` file. Submissions in other formats (e.g., `.docx`, `.txt`) will not be graded. Handwritten solutions are acceptable if scanned clearly. If you prepare your homework in $\text{\LaTeX}$ and ensure a clean, well-formatted layout, you may earn a bonus of up to 20 points.
 - If the homework includes both written and coding parts, two separate upload sections will appear on Moodle: one for the `.pdf` (theoretical part) and one for the `.ipynb` file (coding part). If the homework contains only written or only coding questions, only the relevant section will be available.
 - Do not submit multiple files or a compressed (`.zip`) folder. Only a single, final file will be accepted.
 - Name your file as `name_surname_hw1.pdf` or (e.g., `oktay_altun_hw1.pdf`). Include your name and student ID clearly at the top of your file.
 - **Academic integrity:** Cite all external sources you use (books, lecture notes, online materials, etc.). Plagiarism or unreferenced copying will result in a zero grade and potential disciplinary action.
 - Upload your homework to Moodle before the deadline. Late submissions may not be accepted unless stated otherwise.
 - If deemed necessary, the instructor of the course may conduct interviews related to the assignment and request explanations on how the questions were solved.
- ❗ If you have questions about the assignment, you may ask them in the class WhatsApp group. *If your question includes your solution method, do not post it publicly send it privately to the course assistant.*

Good luck, and submit on time!

1. (50 points)

In linear regression, the least-squares solution

$$\min_w \|Xw - y\|_2^2$$

can behave very differently depending on the geometry of the feature matrix X . In this problem, you will study what happens when we add *redundant* (linearly dependent) features: the parameter vector may no longer be unique, but the predictions can still remain uniquely determined.

a) Let

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \end{bmatrix}, \quad y = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Write the two columns of X as

$$x_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad x_1 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad X = [x_0 \ x_1].$$

Construct two new features

$$x'_2 = 2x_1, \quad x'_3 = 3x_1,$$

and build the matrix

$$\tilde{X} = [x_0 \ x_1 \ x'_2 \ x'_3] \in \mathbb{R}^{3 \times 4}.$$

Show analytically that:

- i) $X^\top X$ is invertible,
- ii) $\tilde{X}^\top \tilde{X}$ is *not* invertible,
- iii) even though the parameters w_2, w_3, w_4 in a linear model with design matrix $\tilde{X}$ are non-unique, the predictions $\tilde{X}w$ that minimize the squared error are unique.

b) Now consider a general setting. Let

$$X = [x_1 \ x_2] \in \mathbb{R}^{N \times 2}$$

have full column rank (i.e., x_1 and x_2 are linearly independent). Define a redundant third feature

$$x_3 = x_1 + x_2, \quad \tilde{X} = [x_1 \ x_2 \ x_3] \in \mathbb{R}^{N \times 3}.$$

- i) Show that the columns of $\tilde{X}$ are linearly dependent and hence $\tilde{X}^\top \tilde{X}$ is not invertible.
- ii) Argue that the least-squares problem $\min_{\tilde{w}} \|\tilde{X}\tilde{w} - y\|_2^2$ has *infinitely many* minimizers $\tilde{w}$.
- iii) Prove that, nevertheless, all least-squares minimizers produce the *same* predictions: for every minimizer $\tilde{w}$ there exists a unique $w \in \mathbb{R}^2$ such that

$$\tilde{X}\tilde{w} = Xw,$$

and the vector Xw is the orthogonal projection of y onto $\text{col}(X)$.

2. (50 points)

a) Consider binary labels $y_i \in \{-1, +1\}$ and the logistic model

$$p(y_i | x_i, w) = \sigma(y_i w^\top x_i),$$

where $\sigma(z) = \frac{1}{1 + e^{-z}}$ is the sigmoid function.

Starting from the joint likelihood

$$p(y_1, \dots, y_N | X, w) = \prod_{i=1}^N p(y_i | x_i, w),$$

derive the *average logistic loss*

$$E_{\text{in}}(w) = \frac{1}{N} \sum_{i=1}^N \ln(1 + e^{-y_i w^\top x_i}).$$

Show every main step: write the log-likelihood, insert the sigmoid, simplify, and obtain $E_{\text{in}}(w)$ as a negative (normalized) log-likelihood.

💡 You may find it useful that

$$\sigma(z) = \frac{1}{1 + e^{-z}} \implies \ln \sigma(z) = -\ln(1 + e^{-z}).$$

b) Compute the gradient of the logistic loss with respect to w :

$$\nabla_w E_{\text{in}}(w).$$

Simplify your answer as much as possible, and express it as a sum over training examples.

c) Let $X \in \mathbb{R}^{N \times d}$ be the data matrix (rows are $x_i^\top$), $y \in \{-1, +1\}^N$ be the label vector, and $w \in \mathbb{R}^d$ be a parameter vector.

Write *vectorized* formulas (no loops) for the following Python functions:

- `sigmoid(z)` that takes a NumPy array `z` and returns $\sigma(z)$ elementwise,
- `logistic_loss(w, X, y)` that returns $E_{\text{in}}(w)$,
- `logistic_grad(w, X, y)` that returns $\nabla_w E_{\text{in}}(w)$.

You do *not* need to implement gradient descent; just give the mathematical vectorized expressions that these functions should compute.